Solutions to Review Questions for Lecture 11

1. a) The prior odds are $0.01/(1-0.01) = 1/99$.
   b) The probability is $P(+) = P(+|disease)P(disease) + P(+|healthy)P(healthy) = 0.01 * 0.8 + 0.99 * 0.05 = 0.0575$.
   c) The data odds are $P(+|disease)/P(+|healthy) = 0.8/0.05 = 16$.
   d) If you interpret the "but" as "and", then $P(disease\ and\ -) = P(disease) * P(-|disease) = 0.01 * 0.2 = 0.002$. If you interpret this as the conditional probability $P(-|disease)$, then the probability is 0.2.
   e) $P(-\ and\ healthy) = P(healthy) * P(-|healthy) = 0.99 * 0.95 = 0.9405$.
   f) Bayes' Theorem gives $P(disease|+) = P(+|disease)P(disease)/P(+) = 0.8*0.01/0.0575 = 0.139$.

2. a) The chance of picking jar A is the chance of landing heads both times, which is $1/2 * 1/2 = 1/4$.
   b) The chance of getting a red marble is $P(jar\ A) * P(red|jar\ A) + P(jar\ B) * P(red|jar\ B) + P(jar\ C) * P(red|jar\ C) = 1/4 * 7/10 + 2/4 * 2/10 + 1/4 * 5/10 = 0.4$.
   c) The chance is $P(jar\ A|red) = P(red|jar\ A) * P(jar\ A)/P(red) = 0.25 * 0.7/0.4 = 0.4375$ by Bayes' Theorem.

3. a) The chance is $P(senior\ and\ math) = P(senior) * P(math|senior) = 0.4 * 0.85 = 0.34$.
   b) The chance is $P(senior|math) = P(senior\ and\ math)/P(math) = 0.34/(0.4 * 0.85 + 0.6 * 0.45) = 0.56$.
   c) The chance is $P(senior)+P(math)-P(senior\ and\ math) = 0.4+0.4*0.85+0.6*0.45-0.4*0.85 = 0.67$.

4. We need $P(defective|pass) = P(pass|defective) * P(defective)/P(pass) = 0.1 * 0.25/(0.1 * 0.25 + 0.95 * 0.75) = 0.033$ by Bayes' Theorem.

5. a) This chance is $3 * P(GGR) = 3 * 1/2 * 1/2 * 1/2 = 3/8$.
   b) This chance is $1 - P(RRR) = 1 - 1/2 * 1/2 * 1/2 = 7/8$.
   c) This chance is $P(green|0\ green)*P(0\ green)+P(green|1\ green)*P(1\ green)+P(green|2\ green)*P(2\ green)+P(green|3\ green)*P(3\ green)$, which evaluates to $0 * 1/8 + 1/3 * 3/8 + 2/3 * 3/8 + 1 * 1/8 = 1/2$.
   d) This means that the box has 2 greens and 1 red, and so the chance is 2/3.
   e) This chance is $P(1\ head|green) = P(green|1\ head)P(1\ head)/P(green)$ by Bayes' Theorem. The chance of exactly 1 head is $3 * P(HTT) = 3/8$ and $P(green|1\ head) = 2/3$. It follows that $P(1\ head|green) = 2/3 * (3/8)/(1/2) = 1/2$.
   f) This chance is $P(\geq 1\ head|green) = P(1\ head|green) + P(2\ head|green) + P(3\ head|green)$. We know that it is impossible to get 3 heads if the marble is green, and so Bayes' Theorem gives the chance as $P(green|1\ head)P(1\ head)/P(green) + P(green|2\ head) * P(2\ head)/P(green) = 2/3 * 3/8/(1/2) + 1/3 * 3/8/(1/2) = 3/4$.
   g) The chance is $P(\geq 1\ tail|green) = P(\geq 1\ tail\ and\ green)/P(green) = P(green)/P(green) = 1$ because it is possible to draw a green marble if and only if there was at least one tail.

6. If you repeat the experiment in question 2 many times and threw out the non-red marbles, then roughly 44% of the marbles remaining would come from jar A. If you repeated the experiment in question 5 many times and only considered the experiments that gave a green marble, then all of the corresponding boxes would have resulted from 3 coin tosses with at least one tail and roughly 75% of the boxes resulted from 3 coin tosses with at least one head.

7. This would be quite a bit lower than 99% because there are many more healthy people who test positive than unhealthy people who test positive.

8. a) The prior odds are $(1/10,000)/(1-1/10,000) = 1/9,999$.
   b) This is $P(+|dangerous)/P(+|not\ dangerous) = 0.99/0.01 = 99$.
   c) This is $99/9,999 = 1/101$.
   d) This is $1/101/(1+1/101) = 1/102$.
   e) Let the data odds be x. The posterior odds would need to be $1 = x/9,999$ for there to be 50% chance, and so $x = 9,999$.

9. Undergoing surgery would result in a 5% chance of death. The chance that the cancer is dangerous is less than 1% from question 8, and so doing nothing would result in a chance of death less than 1%. The problem here is that more than 99% of the people who test positive don't have the dangerous form of the cancer.

10. It might be useful to repeat the test if testing positive twice is much less likely than healthy patients than testing positive once. A single positive test is not that useful however.

11. If you tested many patients and only considered the patients who tested positive, then roughly 1 in 102 of those patients would have the dangerous form of the cancer.